



Export Logistic Management System

- **Team Members:-**

Akhil Rachhadia - 202201005

Sujal Mavadiya - 202201040

Meet Joshi - 202201065

Sahil Vaghasiya - 202201083

Parth Dholariya - 202201085

Normalization Proof

Minimal FD Set and BCNF Proof

1. Seller Table:

- $SID \rightarrow Name$
- $SID \rightarrow Contact$
- $SID \rightarrow Email$
- $SID \rightarrow Address$
- $SID \rightarrow Zip\ Code$

Closure of $SID \Rightarrow \{SID\}^+ = \{SID, Name, Contact, Email, Address, Zip\ Code\}$
Therefore Key = SID .

→All Projected FDs of this Relation have Key on Left side.
→Hence, it is in BCNF Normal Form.

2. Agency Table:

- $AID \rightarrow Name$
- $AID \rightarrow Email$
- $AID \rightarrow Rating$

Closure of $AID \Rightarrow \{AID\}^+ = \{AID, Name, Email, Rating\}$
Therefore Key = AID .

→All Projected FDs of this Relation have Key on Left side.
→Hence, it is in BCNF Normal Form.

3. Agency Location Table:

- $\{AID, Zipcode\} \rightarrow \{Contact\}$
- $\{AID, Zipcode\} \rightarrow \{Address\}$
- Key $\Rightarrow \{AID, Zip\ Code\}$

Closure of $\{AID, Zipcode\} \Rightarrow \{AID, Zipcode\}^+ = \{AID, Zipcode, Contact, Address\}$
Therefore Key = $\{AID, Zipcode\}$.

→All Projected FDs of this Relation have Key on Left side.
→Hence, it is in BCNF Normal Form.

4. Goods Table:

- $GID \rightarrow Name$
- $GID \rightarrow Type$
- $GID \rightarrow PerUnitTax$

Closure of $GID \Rightarrow \{GID\}^+ = \{GID, Name, Type, PerUnitTax\}$.
Therefore Key = GID .

→All Projected FDs of this Relation have Key on Left side.
→Hence, it is in BCNF Normal Form.

5. Have Table:

Key $\Rightarrow \{AID, GID\}$

There is no Projected FD for This relation. Since all attributes together form a Key,
Hence it is in BCNF Normal Form.

6. Charges Table:

- $\{AID, G_Type, Pickup_Location, Delivery_Location\} \rightarrow \{Charge\}$

Closure of $\{AID, G_Type, Pickup_Location, Delivery_Location\} \Rightarrow$
 $\{AID, G_Type, Pickup_Location, Delivery_Location\}^+ = \{AID, G_Type, Pickup_Location, Delivery_Location, Charge\}$

Therefore Key = $\{AID, G_Type, Pickup_Location, Delivery_Location\}$

→All Projected FDs of this Relation have Key on Left side.
→Hence, it is in BCNF Normal Form.

7. Ship Table:

- $Ship_ID \rightarrow Name$
- $Ship_ID \rightarrow G_Type$
- $Ship_ID \rightarrow Availability$
- $Ship_ID \rightarrow AID$
- $Ship_ID \rightarrow Pickup_Location$

- Ship_ID → Delivery_Location
Closure of Ship_ID =>
 {Ship_ID} = {Ship_ID, Name, G_Type, Availability, AID, Pickup_Location, Delivery_Location}
Therefore Key = Ship_ID.

→All Projected FDs of this Relation have Key on Left side.

→Hence, it is in BCNF Normal Form.

8. Invoice Table:

- Inv_no → Inv_Date
- Inv_no → AID
- Inv_no → SID
- Inv_no → CID
- Inv_no → GID
- Inv_no → Qty
- Inv_no → Total_Amount
- Inv_no → Payment_Status
- Inv_no → Order_Status

Closure of Inv_no =>

{Inv_no} = {Inv_no, Inv_Date, AID, SID, CID, GID, Qty, Total_Amount, Payment_Status, Order_Status}

Therefore Key = Inv_no.

→All Projected FDs of this Relation have Key on Left side.

→Hence, it is in BCNF Normal Form.

9. Shipment_Location Table:

- Tracking_ID → Ship_ID
- Tracking_ID → Estimated_arrival_date
- Tracking_ID → Nearest_Country

Closure of Tracking_ID =>

{Tracking_ID} = {Tracking_ID, Ship_ID, Estimated_arrival_date, Nearest_Country}

Therefore Key = Tracking_ID.

→All Projected FDs of this Relation have Key on Left side.

→Hence, it is in BCNF Normal Form.

10. Dockyard Table:

- Dockyard_Code → Name
- Dockyard_Code → Contact
- Dockyard_Code → Email

- Dockyard_Code → Address
 - Dockyard_Code → Zip Code
- Closure of Dockyard_Code =>
 $\{ \text{Dockyard_Code} \}^+ = \{ \text{Dockyard_Code}, \text{Name}, \text{Contact}, \text{Email}, \text{Address}, \text{Zip Code} \}$
 Therefore Key = Dockyard_Code.

→All Projected FDs of this Relation have Key on Left side.

→Hence, it is in BCNF Normal Form.

11. Payment Table:

- Payment_ID → Ship_ID
- Payment_ID → Inv_no
- Payment_ID → Payment_Date
- Payment_ID → Payment_Type
- Payment_ID → Amount_Paid

Closure of Payment_ID =>
 $\{ \text{Payment_ID} \}^+ = \{ \text{Payment_ID}, \text{Ship_ID}, \text{Inv_no}, \text{Payment_Date}, \text{Payment_Type}, \text{Amount_Paid} \}$
 Therefore Key = Payment_ID.

→All Projected FDs of this Relation have Key on Left side.

→Hence, it is in BCNF Normal Form.

12. Zipcode Table:

- Zipcode → City
- Zipcode → State
- Zipcode → Country

Closure of Zipcode => $\{ \text{Zipcode} \}^+ = \{ \text{Zipcode}, \text{City}, \text{State}, \text{Country} \}$
 Therefore Key = Zipcode.

→All Projected FDs of this Relation have Key on Left side.

→Hence, it is in BCNF Normal Form.

13. Consumer Table:

- CID → Name
- CID → Contact
- CID → Email
- CID → Address
- CID → Zip Code

Closure of CID => $\{ \text{CID} \}^+ = \{ \text{CID}, \text{Name}, \text{Contact}, \text{Email}, \text{Address}, \text{Zip Code} \}$
 Therefore Key = CID.

→All Projected FDs of this Relation have Key on Left side.
→Hence, it is in BCNF Normal Form.

14. Seller_Rating Table:

- $\{SID, AID\} \rightarrow \text{Seller_Rating}$
- $\{SID, AID\} \rightarrow \text{Reviews}$

Closure of $\{SID, AID\} \Rightarrow \{SID, AID\}^+ = \{SID, AID, \text{Seller_Rating}, \text{Reviews}\}$
Therefore Key = $\{SID, AID\}$.

→All Projected FDs of this Relation have Key on Left side.
→Hence, it is in BCNF Normal Form.

15. Consumer_Rating Table:

- $\{CID, AID\} \rightarrow \text{Consumer_Rating}$
- $\{CID, AID\} \rightarrow \text{Reviews}$

Closure of $\{CID, AID\} \Rightarrow \{CID, AID\}^+ = \{CID, AID, \text{Consumer_Rating}, \text{Reviews}\}$
Therefore Key = $\{CID, AID\}$.

→All Projected FDs of this Relation have Key on Left side.
→Hence, it is in BCNF Normal Form.